A SECOND CHARACTERISATION OF INJECTIVE MODULES THROUGH BAER'S CRITERION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- Injective module and the extension property for inclusions of submodules. defined earlier in the chapter, source lines 13552-13690, before the counted statement. Supplies the property being characterised.
- Baer's criterion: it suffices to test the extension property on ideals of the ring. recalled and used at proof lines 13703-13720. Reduces the verification to the case of ideals.
- Zorn's lemma applied to partial extensions and the reduction to a cyclic extension. used at proof lines 13720-13750. Produces the maximal counterexample and the single step that has to be verified.

Source reference. Original lemma 0AVF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13683–13683. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 065P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7908–7908. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05CF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6918–6918. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. Let J be an R-module. The following are equivalent*

(1) *J is injective,*
(2) *$\mathrm{Ext}^1_R(R/I, J) = 0$ for every ideal $I \subset R$, and*
(3) *for an ideal $I \subset R$ and module map $I \to J$ there exists an extension $R \to J$.*

Proof. If $I \subset R$ is an ideal, then the short exact sequence $0 \to I \to R \to R/I \to 0$ gives an exact sequence

$$\mathrm{Hom}_R(R, J) \to \mathrm{Hom}_R(I, J) \to \mathrm{Ext}^1_R(R/I, J) \to 0$$

by Algebra, Lemma 065P and the fact that $\mathrm{Ext}^1_R(R, J) = 0$ as R is projective (Algebra, Lemma 05CF). Thus (2) and (3) are equivalent. In this proof we will show that (1) $\Leftrightarrow$ (3) which is known as Baer's criterion.

Assume (1). Given a module map $I \to J$ as in (3) we find the extension $R \to J$ because the map $\mathrm{Hom}_R(R, J) \to \mathrm{Hom}_R(I, J)$ is surjective by definition.

Assume (3). Let $M \subset N$ be an inclusion of R-modules. Let $\varphi : M \to J$ be a homomorphism. We will show that φ extends to N which finishes the proof of the lemma. Consider the set of homomorphisms $\varphi' : M' \to J$ with $M \subset M' \subset N$ and $\varphi'|_M = \varphi$. Define $(M', \varphi') \geq (M'', \varphi'')$ if and only if $M' \supset M''$ and $\varphi'|_{M''} = \varphi''$. If $(M_i, \varphi_i)_{i \in I}$ is a totally ordered collection of such pairs, then we obtain a map $\bigcup_{i \in I} M_i \to J$ defined by $a \in M_i$ maps to $\varphi_i(a)$. Thus Zorn's lemma applies. To conclude we have to show that if the pair (M', φ') is maximal then $M' = N$. In other words, it suffices to show, given any subgroup $M \subset N$, $M \neq N$ and any $\varphi : M \to J$, then we can find $\varphi' : M' \to J$ with $M \subset M' \subset N$ such that (a) the inclusion $M \subset M'$ is strict, and (b) the morphism φ' extends φ.

To prove this, pick $x \in N$, $x \notin M$. Let $I = \{f \in R \mid fx \in M\}$. This is an ideal of R. Define a homomorphism $\psi : I \to J$ by $f \mapsto \varphi(fx)$. Extend to a map $\tilde{\psi} : R \to J$ which is possible by assumption (3). By our choice of I the kernel of $M \oplus R \to J$, $(y, f) \mapsto \varphi(y) + \tilde{\psi}(f)$ contains the kernel of the map $M \oplus R \to N$, $(y, f) \mapsto y + fx$. Hence this homomorphism factors through the image $M' = M + Rx$ and this extends the given homomorphism as desired. $\square$